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Loneliness, social contacts and Internet addiction: A cross-lagged panel study

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ABSTRACT

This study aims to examine the causal priority in the observed empirical relationships between Internet addiction and other psychological problems. A cross-lagged panel survey of 361 college students in Hong Kong was conducted. Results show that excessive and unhealthy Internet use would increase feelings of loneliness over time. Although depression had a moderate and positive bivariate relationship with Internet addiction at each time point, such a relationship was not significant in the cross-lagged analyses. This study also found that online social contacts with friends and family were not an effective alternative for offline social interactions in reducing feelings of loneliness. Furthermore, while an increase in face-to-face contacts could help to reduce symptoms of Internet addiction, this effect may be neutralized by the increase in online social contacts as a result of excessive Internet use. Taken as a whole, findings from the study show a worrisome vicious cycle between loneliness and Internet addiction.

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1. Introduction

Researchers around the world from the domains of communication, psychology, and psychiatry have paid considerable attention in recent years to the rising issue of problematic use of the Internet (PIU), or Internet addiction (IAD) (Griffiths, Miller, Gillespie, & Sparrow, 1999; Tao et al., 2010; Young, 2004; Zhang, Amos, & McDowell, 2008). While the ontological and epidemiological issues surrounding Internet addiction being a clinical pathology are still the subject of many intellectual debates (Czincz & Hechanova, 2009; Frances & Widiger, 2012; Pies, 2009; Widyanto & Griffiths, 2006), social and behavioral scientists generally believe that such a phenomenon indeed exists (Chou, Condron, & Belland, 2005) and that excessive and compulsive Internet use would negatively impact a person's physical, psychological, and social well-being (Davis, 2001; Young, 1998).

Many studies have examined Internet addiction, its social and psychological correlates, and possible treatments (Beard, 2005). While this body of research has identified a number of related factors to the problem, such as depression, loneliness, and social isolation (Ceyhan & Ceyhan, 2008; Chou et al., 2005; Davis, 2001; Young & Roger, 1998), the causal direction of the relationships between these psychosocial problems and Internet addiction have not yet been clearly established (Chou et al., 2005). To fill this void, this study aims to determine the causal priority in the empirical relationship between Internet addiction and Internet users' subjective feelings of depression and loneliness. The effects of online and offline social interactions are also examined.

2. Conceptualizing Internet addiction

Various labels (e.g., compulsive Internet use, pathological Internet use, Internet addiction disorder, Internet usage disorder, and problematic Internet use, etc.) have been adopted to describe the phenomenon of people engaging in excessive and unhealthy use of the Internet. (Widyanto & Griffiths, 2006). It has been widely recognized as a significant mental health issue worldwide. In 2012, the American Psychiatric Association recommended including "Internet Use Disorder" for further study in Section III of the DSM-5. In Korea, Internet addiction has been deemed a national health concern estimated to affect up to 30% of Internet users under 18 (Fackler, 2007; Ha et al., 2007). In China, more than 10% of adolescent Internet users were identified as Internet addicts (Block, 2008; Wu & Zhu, 2004), and the Chinese government went so far as sponsoring various treatment clinics to deal with this new "addiction". In Taiwan, nearly 6% of college students were believed to be affected by problematic Internet use (Chou & Hsiao, 2000).

In general, Internet addiction has been conceptualized as a behavioral control problem. It refers to Internet users' inability to control their use of the medium, which in turn might cause one's marked distress and functional impairment in daily life (Shek, Sun, & Yu, 2013). This condition often shares similar epidemiological properties and consequences as some classified addictions

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such as substance dependency (Brenner, 1997) and impulse control disorders (Young, 1998, 1999, 2004, 2009). To this extent, a number of diagnostic criteria have been proposed and tested (Beard, 2005; Brenner, 1997; Byun et al., 2009; Chou et al., 2005; Demetrovics, Szeredi, & Rozsa, 2008; Ha et al., 2007; Young, 1999). However, methods of clinically assessing Internet addiction disorder remain limited and inconsistent (Chou et al., 2005; Dell'Osso, Altamura, Allen, Marazziti, & Hollander, 2006). A number of epidemiological and theoretical issues related to the proposed disorder have not been fully resolved through empirical research; for example, internet addiction may be a symptom resulted from other existing psychiatric disorders, such as depression, social anxiety, and generalized obsessive and compulsive disorder (Beard, 2005; Ha et al., 2007; Pies, 2009).

Adding to the complexity in conceptualizing Internet addiction as a psychiatric condition, the very notion of "Internet use" has also changed dramatically since researchers began to investigate this problem almost 20 years ago. Early forms of the Internet were simply channels of electronic communication. Users mainly browsed information displayed on static webpages and/or sent text-based messages to each other. As computing and network technologies advance, however, the Internet is no longer just a communication channel. People nowadays conduct more and more of their routine social and professional activities online. The Internet is now quite literally a common thread that weaves the fabric of our lives. Furthermore, mobile devices such as smartphones and wearable computers allow people to stay connected wherever and whenever. A growing number of Internet users spend nearly as much, if not more time, online as they do offline. Thus, it is problematic to use the amount of time spent online or a general dependency on the Internet technology as primary indicators of problematic use. It is also difficult to determine to what aspects of the Internet people are "addicted".

Despite a lack of consensus on the conceptualization and assessment of Internet addiction and the ambiguity in defining "Internet use", most social scientists agree that Internet addiction, as a general social phenomenon, indeed exists (Chou et al., 2005). Those who are affected by this problem often spend an excessive amount of time on the Internet, which adversely affects their offline lives; they may develop a preoccupation with online activities, feel the need to escape into cyberspace, and express an increased amount of irritability when trying to cut back their Internet use (Dell'Osso et al., 2006). These individuals may also experience a number of functional impairments as a result of using the Internet such as marital or family strife, job loss or decreased job productivity, and legal difficulties or school failure (Chou & Hsiao, 2000).

In the present study, we are not interested in engaging the debate surrounding Internet addiction in a clinical sense, but rather as a widely observed social phenomenon. As such, we will use the term Internet addiction throughout this paper to describe a general psychological condition in which an individual's mental and emotional states are adversely affected by the overuse of the medium (Beard, 2005).

3. Internet addiction and psychosocial problems

Numerous studies from around the world found Internet addiction to be positively associated with other psychosocial problems, such as feelings of depression and loneliness (Beard, 2005; Chou et al., 2005; Ha et al., 2007; Tokunaga & Rains, 2010). Morahan-Martin and Schumacher (2000) found that 8% of their sample of undergraduate students in the United States were engaging in pathological Internet use, and were experiencing a greater degree of loneliness than their non-addicted counterparts. In a cross-sectional study, Kim, LaRose, and Peng (2009) identified loneliness to be the cause, as well as the effect, of problematic Internet use among American college students. Yen, Ko, Yen, Chang, and Cheng (2009) recently surveyed 8941 adolescents in Taiwan; they found depression and a lack of family contact to be discriminating factors for Internet addiction. Similar patterns were also found in China (Huang & Deng, 2009; Liu, Xu, & Hu, 2009), South Korea (Byun et al., 2009; Park, 2009), Norway (Bakken, Wenzel, Götestam, Johansson, & Øren, 2009), and Iran (Ghassemzadeh, Shahraray, & Moradi, 2008). Most recently, Tokunaga and Rains (2010) conducted a meta-analysis of 94 studies from 22 different countries and found moderate and consistent links between loneliness, depression, and problematic Internet use.

Davis (2001) posited that existing psychopathological disorders, such as depression, loneliness, social anxiety, and substance dependence were necessary elements in the etiology of pathological Internet use. Specifically, people with psychosocial dispositions such as depression and loneliness would be prone to hold maladaptive cognitions (e.g., they would only feel good on the Internet and/or the offline world is awful). Therefore, those who feel depressed and lonely would be especially vulnerable to Internet addiction. Extending from Davis's cognitive–behavioral model of problematic Internet use, Caplan (2003) further argued that individuals with psychosocial problems (e.g., depression and loneliness) were more likely to perceive themselves as unskilled in social competence. They would prefer computer-mediated interactions rather than face-to-face communication because the anonymity and a lack of non-verbal cues in computer-mediated communication would make online social interaction less threatening. As a result, these individuals would be inclined to perceive themselves as more sophisticated or successful in online social interactions. This preference would in turn lead to excessive and compulsive use of the Internet. Based on the theoretical reasoning outlined above, we would predict that:

H1. Feelings of depression will lead to higher levels of Internet addiction.

H2. Feelings of loneliness will lead to higher levels of Internet addiction.

More interestingly, if depression and loneliness drive people to use the Internet more often, will such an increase of usage provide a relief for these negative psychological feelings? From a social network and social capital perspective, Wellman et al. (1996) argued that the Internet could strengthen previously formed social relations because it would add effective means of communication and social contact. Supporting this view, Howard, Raine, and Jones (2002) found that, after controlling for basic socio-demographics, those who had gone online had a 24% greater likelihood of saying that they knew people who could be helpers in times of need than those who had never gone online. A variety of early studies (Constant, Sproull, & Kiesler, 1996; Rice, 1999; Sproull & Kiesler, 1991) have also found that computer-mediated communication extended possibilities for social connection by crossing time, geography and space, allowing people with different backgrounds to share interests and to come together in a virtual world. From this perspective, it seems that Internet use may foster virtual, but still real and meaningful, social interactions that would in turn reduce feelings of loneliness. This suggests that while feelings of depression and loneliness may lead to Internet addiction in the short run, as predicted in H1 and H2, the reduction of these feelings associated with greater virtual social connections may therefore also reduce Internet addiction over time. Based on this line of reasoning, we propose the following hypothesis and research question:

H3. Online social contacts will help reduce the effects of loneliness and depression on Internet addiction.

Despite the optimistic view that online social contacts might be socially beneficial, some evidence suggest that if Internet use is excessive, it may lead to increased social isolation from a decrease of face-to-face social interactions. Young and Roger (1998) suggested that increased levels of social isolation and depression may result from excessive Internet use, rather than being the cause of Internet addiction. Young (1998) found that excessive use of the Internet led to family problems because some users indulged in cyber-affairs or extramarital relationships that would destabilize real-life social relationships. Kim et al. (2009), through a path analysis of a cross-sectional survey, showed that individuals who were lonely or lacked good social skills could develop strong compulsive Internet use behaviors. Such compulsive Internet use may further prevent users from forming healthy social interactions and thus lead to more loneliness. Furthermore, from the perspective of displacement theory (Kraut et al., 1998; Nie, Hillygus, & Erbring, 2002), one could also argue that the Internet may undermine people's social development by taking away time that would otherwise be spent with family and friends. The following hypothesis and research question can thus be derived:

H4. Internet addiction will lead to a reduction in the frequency of face-to-face social contacts, and therefore lead to an increase in feelings of depression and loneliness.

4. Method

A two-wave panel survey was conducted. Students in two major public universities in Hong Kong were invited to fill out two online questionnaires four months apart. Each participant received a HK$50 cash coupon from a local supermarket (approximately US$6.5) for completing the first survey and another HK$50 upon completion of the follow-up questionnaire. The second-wave questionnaire was nearly identical to the first one.

5. Sample

An invitation email for the survey was distributed through the student unions of the two universities. A total of 636 participants completed the first wave of questionnaires within a period of 2 weeks. A follow-up email was sent to these 636 participants after four months, inviting them to participate in the second-wave survey. A reminder was sent out one week later. In total, 361 of the 636 participants completed both questionnaires. Among the participants, 51.7% of them were males, 37.6% of them were females, while about 10% of them did not report their sex. The mean age of the participants was 21.63 years (SD = 2.60), ranging from 18 to 37. Overall, the sample had a high level of Internet usage. 89% of participants reported daily Internet use, and 62% of them said that they would spend three or more hours online every day.

6. Questionnaire development and measurement

Over 90% of the population in Hong Kong is ethnically Chinese. However, being a British colony for nearly a century, the majority of educated individuals in Hong Kong are bilingual. Both Chinese and English are official languages, and all public documents published by the Government are in both languages. English is also the medium of instruction at the two universities from which our sample was drawn. Given this situation, a bilingual questionnaire was used, with each item in the survey presented in both English and Chinese. Pilot tests, as well as an independent back-translation, were conducted to ensure accuracy of the translations.

6.1. Level of Internet addiction

While a number of diagnostic assessments are available for measuring Internet addiction or problematic Internet use (cf. Beard, 2005; Shapira et al., 2003), this study adapted Young (1998) 20-item Internet Addiction Test (IAT) because of its wide adoption by academic researchers and medical professionals, particularly in Asia. Examples of the questions include: "how often do you find that you stay online longer than you intended"; "how often do you neglect household chores to spend more time online"; and "how often do you prefer the excitement of the internet to intimacy with your partner". All items were measured on a 7-point scale. An average score serves as the final indicator.

6.2. Feelings of depression

A shortened version of Hamilton (1960) depression rating scale (HDRS) was used in the present study to gauge participants' self-reported feelings of depression. This scale included 12 items. Participants were asked the report the extent to which they experience symptoms of depression such as feelings of guilt, depressive mood and insomnia. Created for the purpose of clinically assessing depression, the HDRS is a checklist. However, in this study we are primarily interested in evaluating a persons' subjective feelings of depression rather than clinically screening for depression, the items in this scale were measured on a 7-point Likert scale. The average score of all items was used as an indicator for depression.

6.3. Loneliness

Participants' subjective feelings of loneliness were assessed by Russell (1996) 20-item UCLA loneliness scale, measured on a 7-point scale. The average scores were used in all subsequent analyses. Some examples of the items from the scale include: "I am unhappy doing so many things alone"; "I have nobody to talk to"; and "I lack companionship".

6.4. Social contact

An individual's frequency of being in contact with family and friends, both online and offline, were measured. In each wave of the survey, participants were asked, using a 7-point scale, how often they had contacted family and friends face-to-face, via instant messengers, via email, and via social networking sites in the previous month. A summed score of individuals' face-to-face contacts with family and friends was used to gauge offline social interactions, and the summed score of contacts via other means was used to measure online social contact.

Table 1 presents the descriptive statistics of key variables in both waves, while all bivariate correlations are presented in Table 2.

Table 1
Descriptive statistics of key measures.

Variables	1st Wave		2nd Wave	
	Mean (SD)	α	Mean (SD)	α
Depression	3.11 (1.01)	.88	3.09 (.94)	.87
Loneliness	3.98 (.63)	.71	3.84 (.63)	.77
e-Contact (sum of 6 items)	22.80 (6.64)	n/a	22.68 (6.04)	n/a
F2F contact (sum 2 items)	11.64 (1.94)	n/a	11.99 (2.02)	n/a
Internet addiction	3.44 (.93)	.91	3.37 (.98)	.92

7. Data analysis

The present study adopted a classic two-wave cross-lagged panel design (Campbell, 1963; Kenny, 1973). A path analysis approach to analyzing cross-lagged panel data as suggested by Kessler and Greenberg (1981) was employed to test our hypotheses. Specifically, we were interested in evaluating the relative strength of the cross-lagged relationships between the set of variables *X* and *Y* measured in time 1 and time 2 in a cross-lagged panel structural model (Fig. 1). To ensure consistency in our interpretations of the results we used the standardized z-scores of the key variables in all analyses. All path models were tested in AMOS 19.

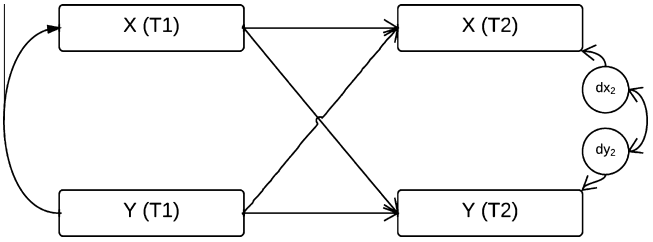


Fig. 1. A two-wave two-variable cross-lagged panel model.

8. Results

8.1. Depression and symptoms of Internet addiction

The first hypothesis (H1) predicted that individuals' subjective feelings of depression would lead to symptoms of Internet addiction. As can be seen from Table 1, the bivariate correlations between depression and Internet addiction at both time 1 and time 2, and all cross-lagged correlations between the two variables were significant and positive, suggesting that there was indeed a positive relationship between feelings of depression and symptoms of Internet addiction. However, a test of the cross-lagged structural model revealed that once the stability effect (i.e. the influence of a measure at time 1 on the same measure at time 2) is controlled for, the influence of depression at time 1 on Internet addiction at time 2 became non-significant, so did the influence of Internet addiction at time 1 on depression at time 2. This finding indicates that the relationship between depression and Internet addiction might have been caused by other factors (see Fig. 2). As such, we did not find support for H1, and depression was excluded from the subsequent hypotheses testing.

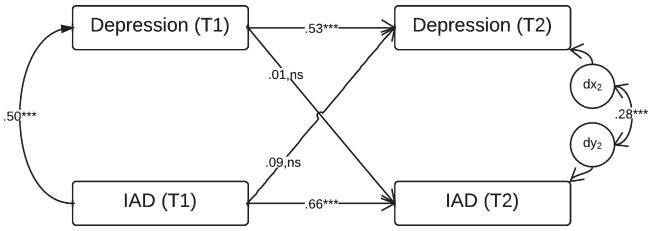


Fig. 2. Standardized regression coefficients in a cross-lagged panel structural model testing the influence of depression on Internet addiction. ***Coefficient is significant at the 0.001 level. n.s. Non-significant.

8.2. Loneliness and symptoms of Internet addition

Our second hypothesis (H2) concerns the relationship between a person's subjective feelings of loneliness and Internet addiction. We expected that feelings of loneliness would not only be positively associated with, but also a causal factor leading to symptoms of Internet addiction. Simple bivariate correlational analyses (see Table 2) showed moderate and positive relationships between loneliness and Internet addiction at each time point. However, after controlling the stability effects (i.e., the influence of loneliness at time 1 on loneliness at time 2 and the influence of IAD at time 1 on IAD at time 2), the cross-lagged effect of loneliness at time 1 on IAD at time 2 faded, but the significant effect from Internet addiction at time 1 on loneliness at time 2 remained (see Fig. 3). This suggests that Internet addiction may be a cause of loneliness, which is the opposite of what we have predicted. H2 is not supported.

8.3. Social contacts and Internet addiction

Hypothesis 3 (H3) predicted that individuals would seek comfort from online social contacts when they feel lonely and depressed, which will in turn reduce symptoms of Internet addiction. We found that the frequency of individuals' online social contacts with family and friends in the months prior to each wave of surveys had a positive correlation with symptoms of Internet addiction at both time points. This is opposite to what we had expected. H3 was not supported.

Table 2
Bivariate correlations between key measures.

	IAD (T1)	IAD (T2)	Depression (T1)	Depression (T2)	Loneliness (T1)	Loneliness (T2)	e-Contact (T1)	e-Contact (T2)	F2F-Contact (T1)	F2F-Contact (T2)
IAD (T1	1									
IAD (T2)	.661**	1								
Depression (T1)	.498**	.344**	1							
Depression (T2)	.352**	.406**	.571**	1						
Loneliness (T1)	.467**	.280**	.639**	.437**	1					
Loneliness (T2)	.343**	.351**	.416**	.622**	.449**	1				
e-Contact (T1)	.133**	.154**	−.019	−.044	−.055	−.067	1			
e-Contact (T2)	.043	.205**	.008	.018	−.095	-.060	.468**	1		
F2F-contact (T1)	−.056	−.008	−.141**	−.124*	−.115*	−.083	.293*	.197**	1	
F2F-contact (T2)	−.025	−.054	−.178**	−.168**	−.150*	−.121	.156	.368**	.399**	1

* Correlation is significant at the 0.05 level (2-tailed).
** Correlation is significant at the 0.01 level (2-tailed).

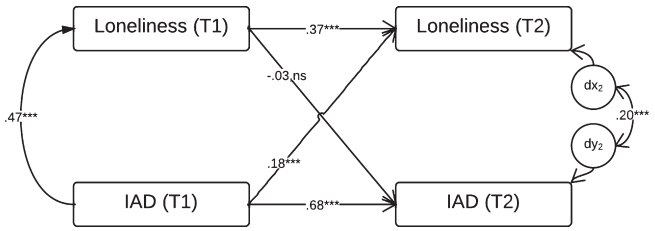


Fig. 3. Standardized regression coefficients in a cross-lagged panel structural model testing the influence of loneliness on Internet addiction. ***Coefficient is significant at the 0.001 level. n.s. Non-significant.

We predicted in Hypothesis 4 (H4) that Internet addiction would have a negative impact on the frequency of an individual's offline social contacts with friends and family, which in turn would lead to increased feelings of loneliness and depression. As can be seen in Table 2, the level of Internet addiction at time 1 was not related to participants' face-to-face contacts at a later time. However, we found that participants' face-to-face social contacts with family and friends in the month prior to the second wave of survey had a significantly negative correlation with symptoms of Internet addiction at time 2. H4 is partially supported.

To further examine the effects of social contacts on Internet addiction and psychological wellbeing, a series of structural models were tested. In both waves of the survey, we asked the participants to report their online and offline contacts with family and friends in the month prior to filling out the questionnaire. Reflecting a temporal order, the social contacts variables were given causal priority. Fig. 4 presents the best-fit model ($\chi^2 = 8.442$, $df = 13$, $p > .80$, $NFI = .987$, $RMSEA = .00$). As it can be seen, offline social contacts reduced participants' feelings of loneliness and symptoms of Internet addiction, but online social contacts increased Internet addiction. Furthermore, what is the most interesting is that while offline social contacts were not correlated with IAD at the bivariate level, the negative relationship between the two became significant once the influence of online social contacts were controlled for in the structural model. This finding suggests that while an increase in face-to-face social contacts would help reducing Internet addiction, this effect might be neutralized by a concurrent increase in the frequency of online social contacts.

9. Discussion

This study conducted a cross-lagged panel survey to establish the causal order in the empirical relationships between subjective feelings of loneliness and depression, and symptoms of Internet addiction.

Our study makes several notable contributions to Internet addiction research. First, the hypotheses in this study were derived from a cognitive behavioral perspective of Internet addiction (Davis, 2001). From this view, pathological Internet use is a maladaptive behavior aimed to reduce negative feelings such as loneliness, depression and social anxiety. Therefore Internet addiction would be the result, instead of the cause of, other psychological problems. As it was found in many previous studies, our analyses first show that depression had a moderate relationship with Internet addiction. However, this relationship seems to have been an artifact of a third variable problem. The cross-lagged panel analyses further revealed that Internet addiction had the causal priority in its empirical relationship with loneliness, suggesting that excessive Internet use would increase feelings of loneliness over time, instead of the opposite. As such, while our study failed to support the cognitive behavioral perspective, it provided evidence for Young and Roger's (1998) view of Internet addiction as a cause of other psychosocial problems. Although its findings are to be validated by future studies, this study is a first attempt to test the causal links between Internet addiction and psychosocial problems of loneliness and depression explicitly.

Second, our findings challenge the assumption that online communications such as social media can effectively provide the social needs people otherwise would fulfill with face-to-face interactions. In this study, we found that while a person's offline social contacts with family and friends might have helped reducing symptoms of Internet addiction, an increase in online social contacts seemed to have neutralized this effect. Moreover, online social contacts did not reduce feelings of loneliness, but loneliness had a negative impact on offline social contacts. This seems to indicate that online social contacts might not be an effective replacement for offline interactions. When we interpret this finding together with the earlier finding that Internet addiction would lead to an increase in loneliness, a worrisome vicious cycle between loneliness and Internet addiction becomes apparent.

Several limitations to this study are worth noting. First, both loneliness and depression are complex psychological conditions extensively studied by previous research (Cabello et al., 2012; Heinrich & Gullone, 2006; McWhirter, 1990). Their correlates may include neurological, psychological, developmental, social, and cultural factors. Driven by a specific goal to examine the causal direction in an empirical relationship found in previous studies, this study has a rather narrow scope, focusing only on Internet addiction as being one of the possible contributors to the subjective feeling of loneliness and depression. As such, this study might

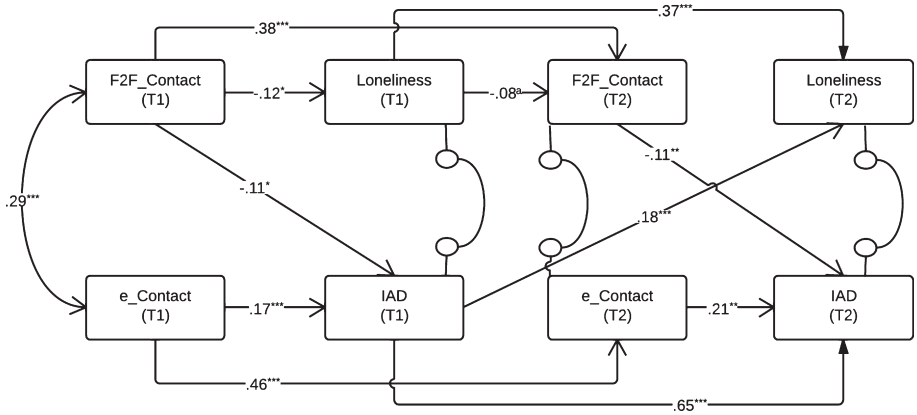


Fig. 4. Standardized regression coefficients in a structural model testing the influence of online and offline social contacts on loneliness and Internet addiction. $\chi^2 = 8.442$, $df = 13$, $p > .80$, NFI = .987, RMSEA = .00. a. Coefficient is marginally significant at the .10 level. *Coefficient is significant at the .05 level. **Coefficient is significant at the .01 level. ***Coefficient is significant at the 0.001 level.

have overlooked the influence of some other factors, such as personality, age, gender, family environment, and education. More sophisticated theoretical models should be developed to address this issue in future studies.

Additionally, as it was alluded earlier in this paper, the Internet is no longer a communication channel through which individuals seek and exchange information. Today, almost all human activities, be it social or professional, could involve the use of the Internet. Different online activities may take place in different social contexts and may have different impacts on people's psychological wellbeing. The existing empirical research on Internet addiction has primarily been informed and driven by theories developed in the fields of clinical psychology. As such, the effects of media content and communication contexts on psychological processes, a core subject of media and communication research, have been generally ignored in this body of literature. For example, communication scholars conceptualized different media channels based on their varying ability to transmit and convey social information (Daft & Lengel, 1984). Some media channels (e.g., videoconference) are more effective in sending information that would facilitate social and emotional development than others (e.g., text-messaging). This might have an impact on Internet addiction. Additionally, over fifty years of media effects research accumulated a rich body of theoretical knowledge about the harmful psychological effects of viewing certain types of media content (Bryant & Oliver, 2009). Yet research on Internet addiction has largely ignored the effects of media content thus far. Clearly, an integration of theories from media and communication studies and behavioral sciences is much needed to extend understandings about Internet addiction.

Methodologically, the time span between the two waves of surveys might have been too short to detect a stable change in Internet addiction and psychosocial problems over time. Furthermore, with only two time points in the cross-lagged model, it is difficult to examine the influence of social contact as a mediator. In the present study, we only used Young (1998) Internet Addiction Test (IAT) as an indicator of Internet addiction. While this scale has been widely adopted by researchers around the world, it would be worthwhile to triangulate our findings with other available measures of problematic Internet use. Although the present study is not interested in clinically diagnosing depression and loneliness, our findings about these two conditions might have been limited because of the use of self-reported measures.

Finally, we had to use a convenient sample of college students in the present study due to constraints of both financial and human resources. This sample was comprised purely of students from Hong Kong, likely representing a collectivistic culture. Differences in culture may affect the relationships between Internet addiction, loneliness, and depression (Anderson, 1999). The participants' age may also play a role. For example, a recent study found that online communication is effective in reducing feelings of loneliness among other adults (Cotten, Anderson, & McCullough, 2013). However, notwithstanding the obvious limitations on external validity, we trust that our findings are robust and internally consistent for the research purpose, given the relative homogeneity in the target population's Internet usage and demographics.

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